Tín Hiệu và Hệ Thống

Chương 3: Biểu diễn hệ thống tuyến tính bất biến trong miền tần số (Phần 4)

VIETNAM NATIONAL UNIVERSITY, HANOI
VNU UNIVERSITY OF ENGINEERING & TECHNOLOGY

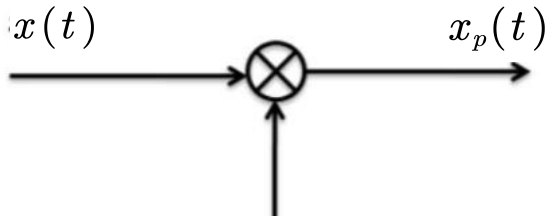
FACULTY OF ELECTRONICS
& TELECOMMUNICATIONS

LẤY MẪU TÍN HIỆU TRONG MIỀN THỜI GIAN

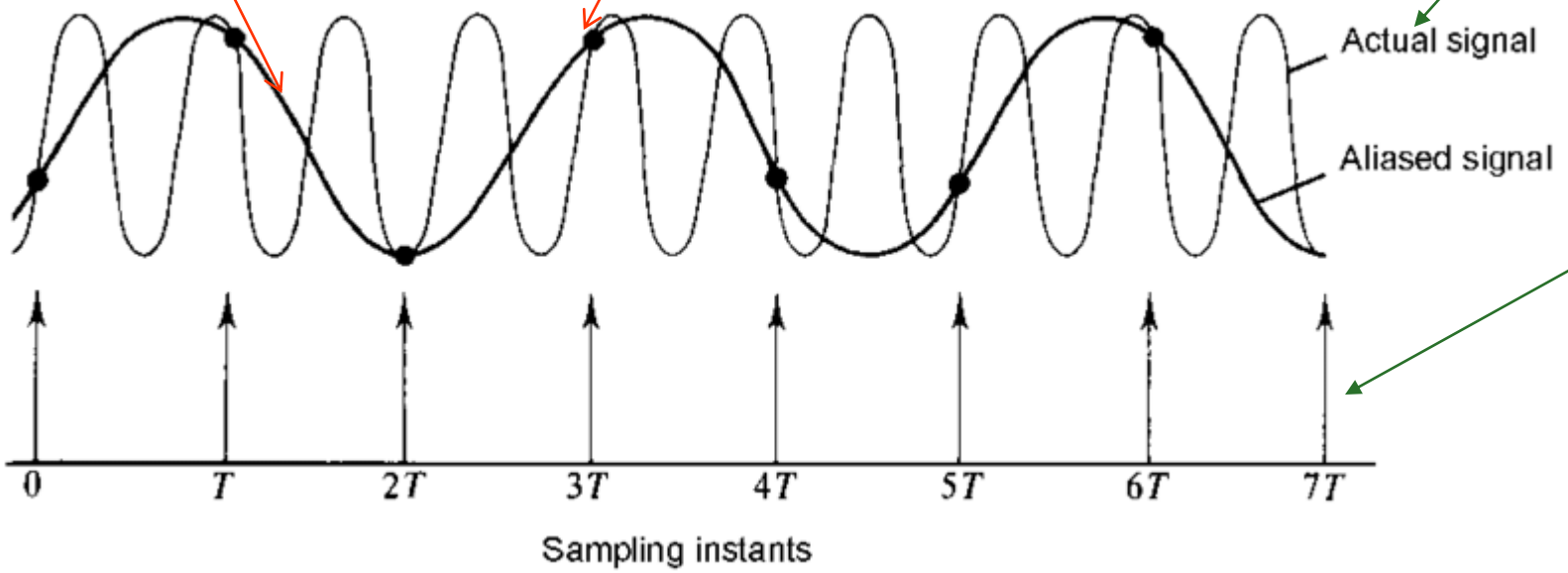
Việc **lấy mẫu dưới** có thể gây nhầm lẫn
Ở đây, **tín hiệu sau lấy mẫu** gợi ý một tần số khác cho tín hiệu **được lấy mẫu**

Dấu chấm thể hiện dữ liệu **sau lấy mẫu**

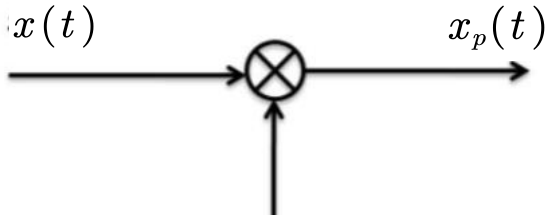
T: chu kì lấy mẫu.



$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$



LẤY MẪU TÍN HIỆU TRONG MIỀN THỜI GIAN



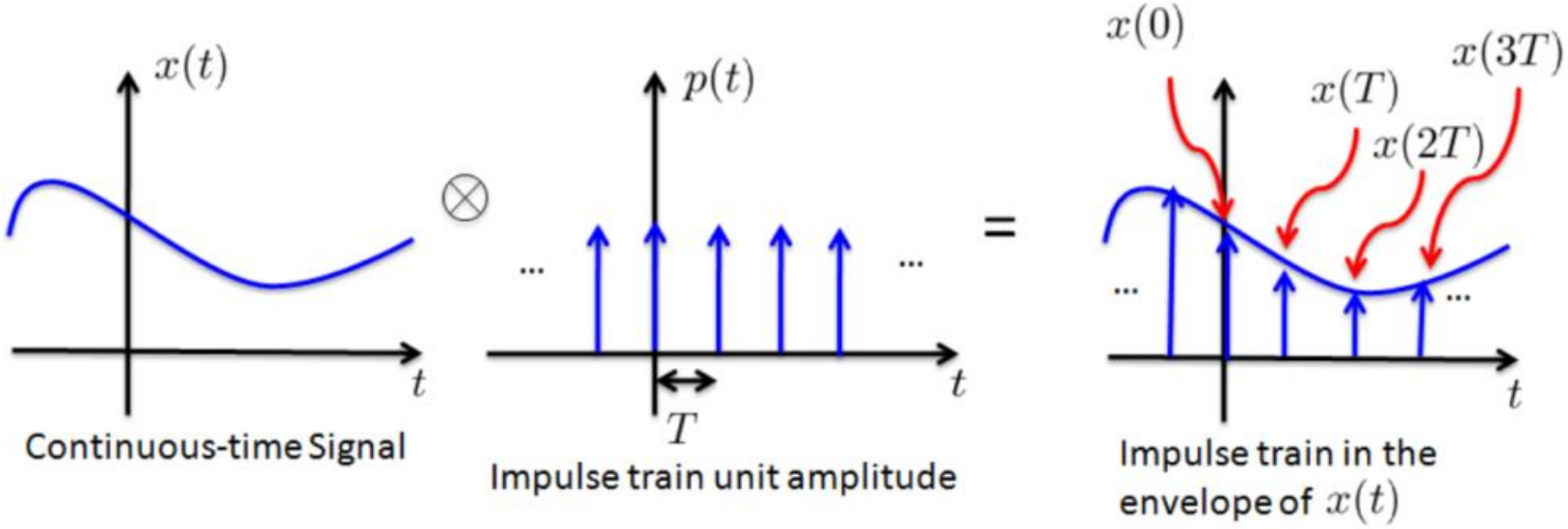
$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

T: chu kì lấy mẫu.

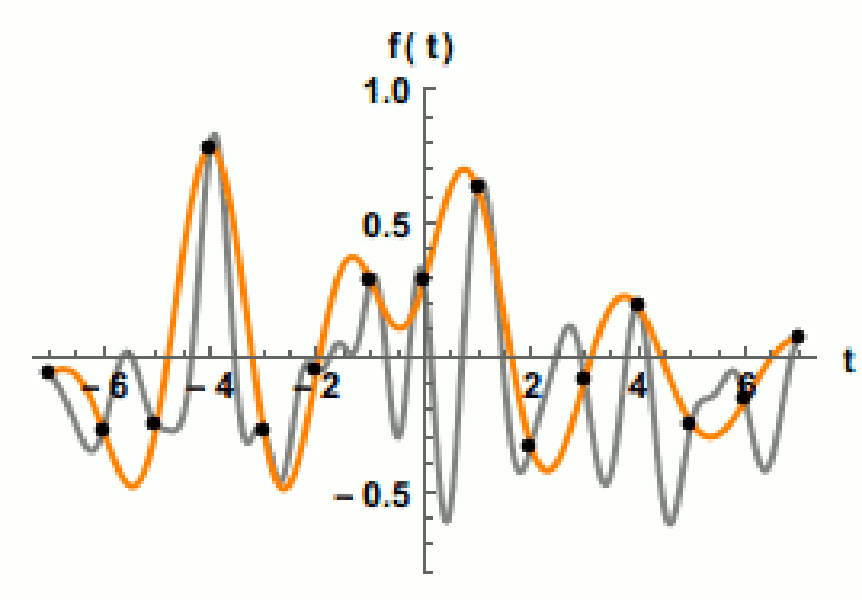
$$x_p(t) = x(t)p(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$= \sum_{n=-\infty}^{\infty} x(t)\delta(t - nT)$$

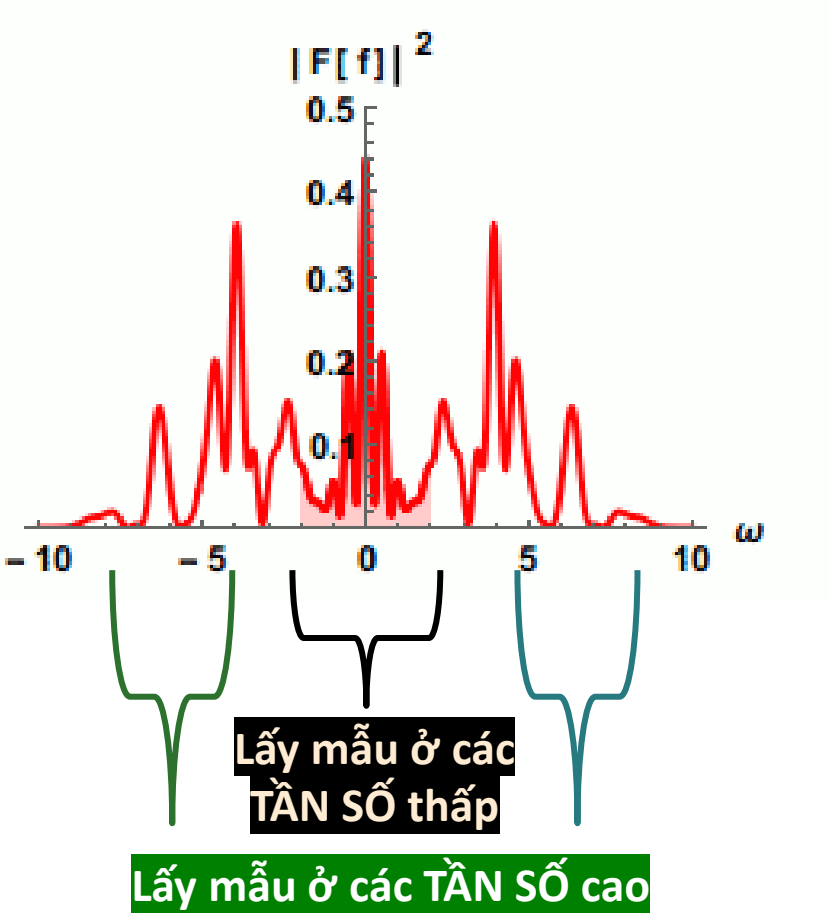
$$= \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT).$$



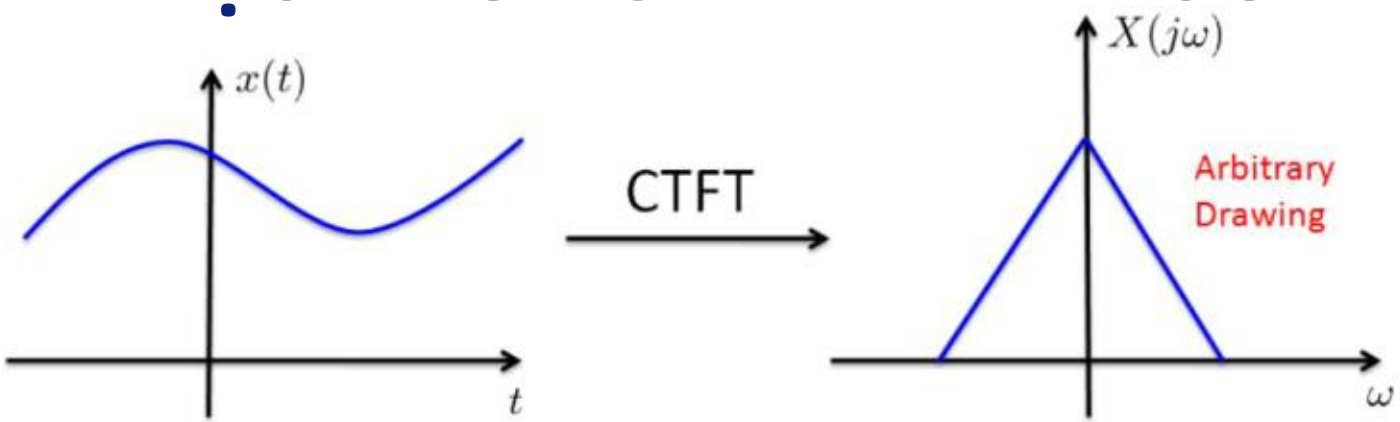
LẤY MẪU TÍN HIỆU TRONG MIỀN THỜI GIAN

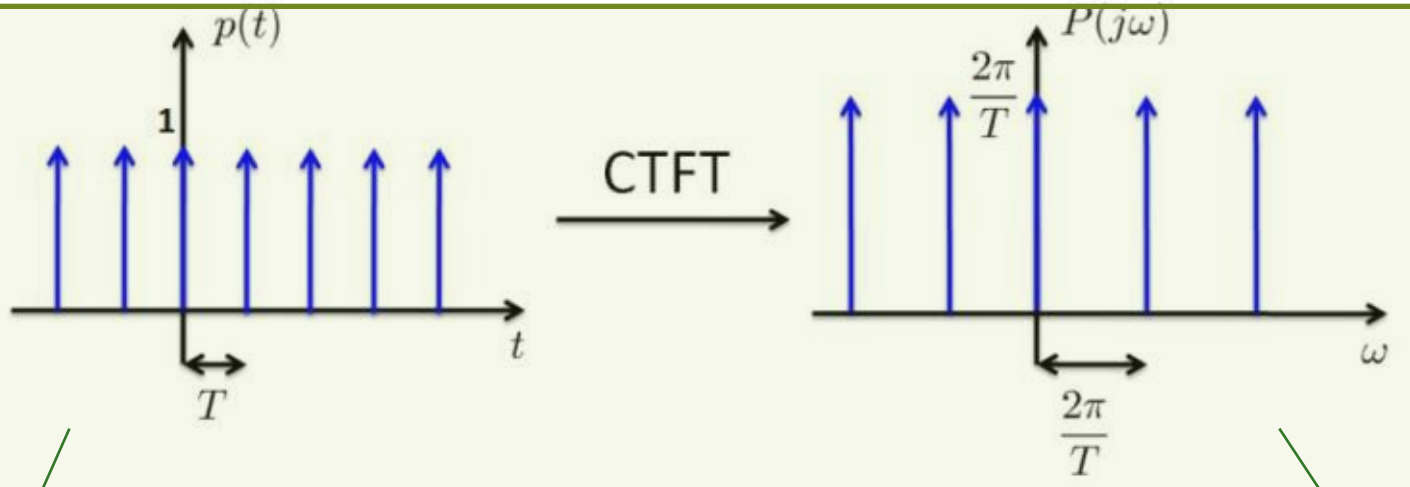


Phổ tần số của tín hiệu f(t)



LẤY MẪU TÍN HIỆU TRONG MIỀN TẦN SỐ



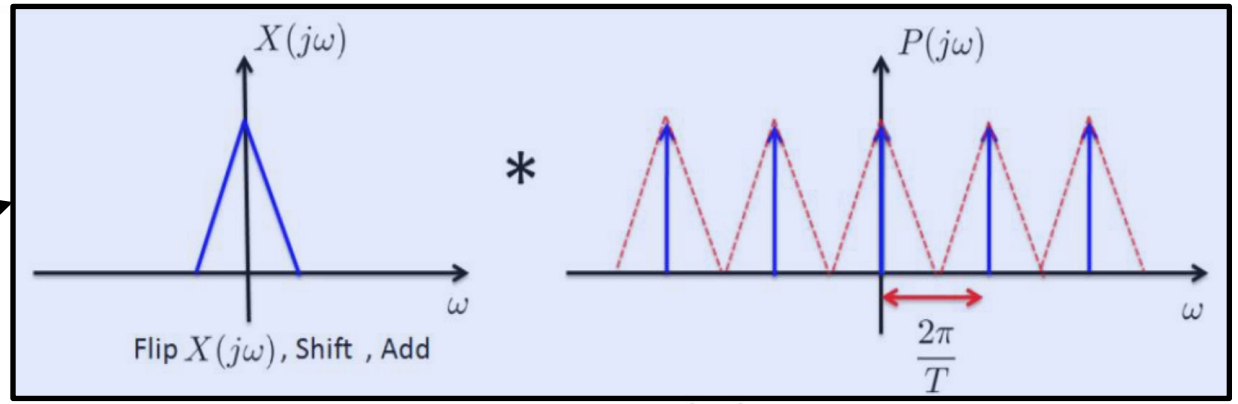


$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \overset{FT}{\longleftrightarrow} \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{T}\right) = P(j\omega)$$

LẤY MẪU TÍN HIỆU TRONG MIỀN TẦN SỐ

$$x_p(t) = x(t)p(t)$$

$$\Rightarrow X_p(j\omega) = \frac{1}{2\pi} X(j\omega) * P(j\omega)$$

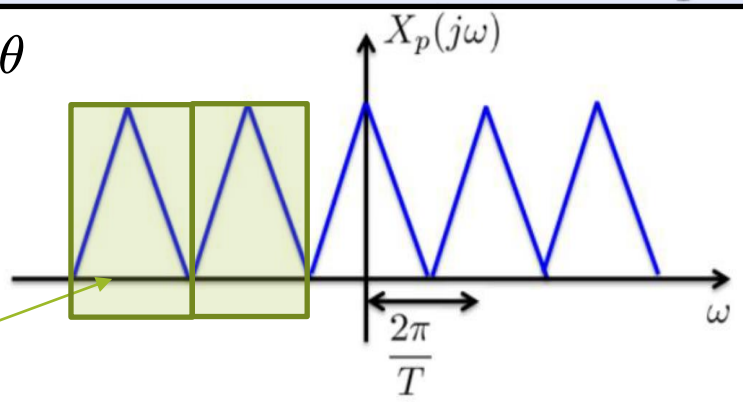


$$X_p(j\omega) = \frac{1}{2\pi}\boxed{X(j\omega) * P(j\omega)} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\theta)P(j(\omega-\theta))\,d\theta$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\theta)\left[\frac{2\pi}{T}\sum_{k=-\infty}^{\infty}\delta\left(\omega-\theta-\frac{2\pi k}{T}\right)\right]d\theta$$

$$= \frac{1}{T}\sum_{k=-\infty}^{\infty}\left[\int_{-\infty}^{\infty} X(j\theta)\delta\left(\omega-\theta-\frac{2\pi k}{T}\right)d\theta\right]$$

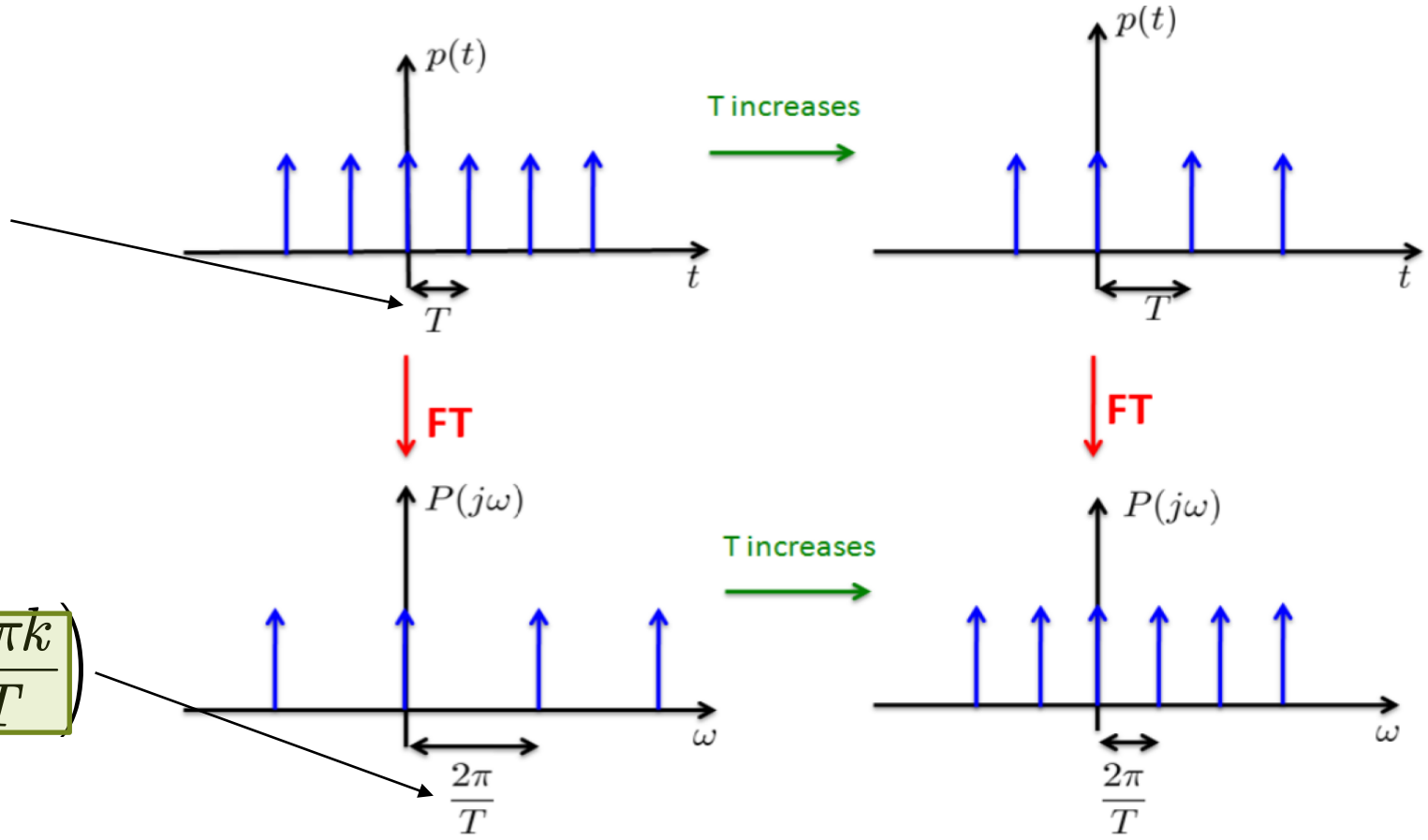
$$= \frac{1}{T}\sum_{k=-\infty}^{\infty}\boxed{X\left(j\left(\omega-\frac{2\pi k}{T}\right)\right)}$$



LẤY MẪU TÍN HIỆU TRONG MIỀN TẦN SỐ

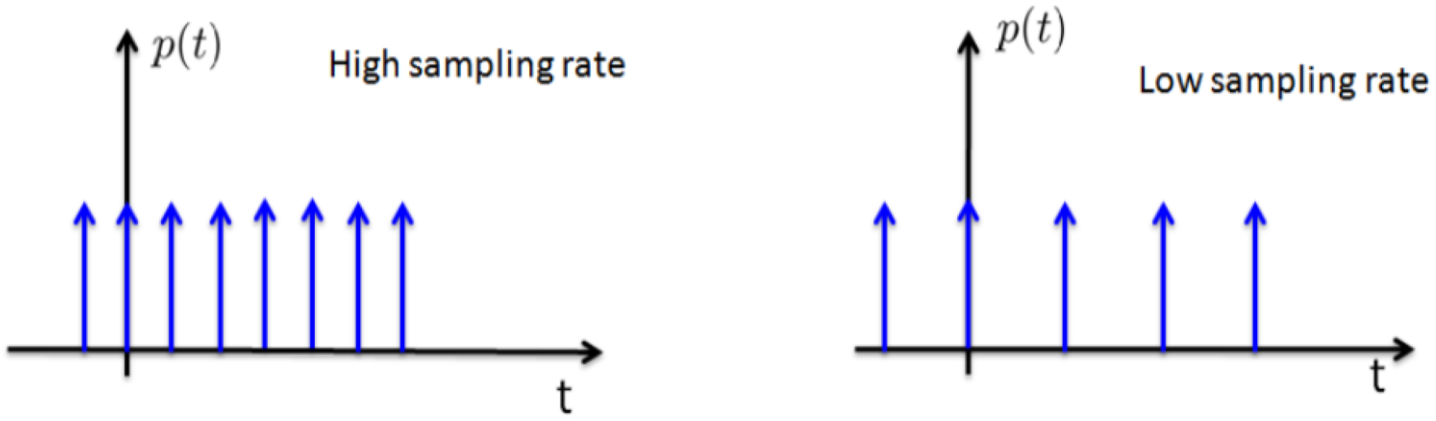
$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$$

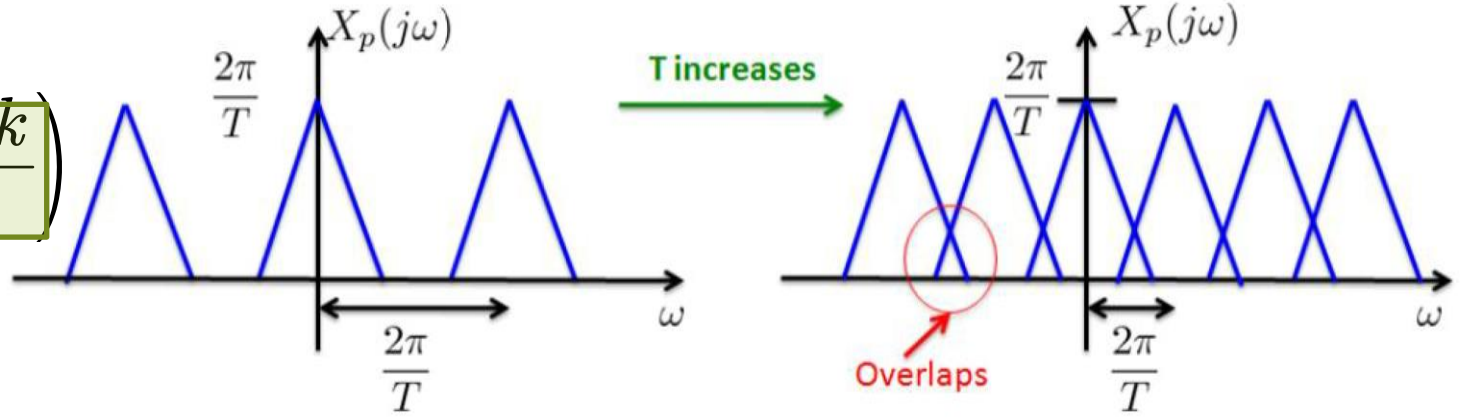


LẤY MẪU TÍN HIỆU TRONG MIỀN TẦN SỐ

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$



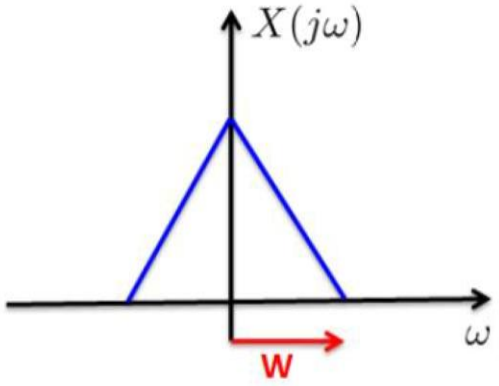
$$P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{T}\right)$$

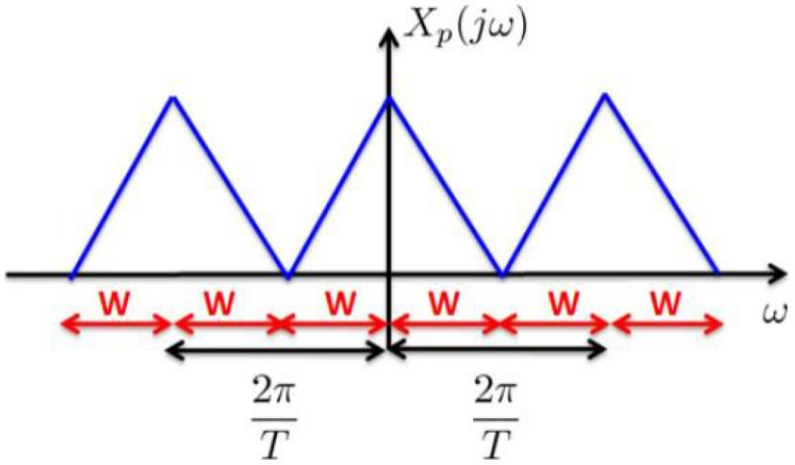


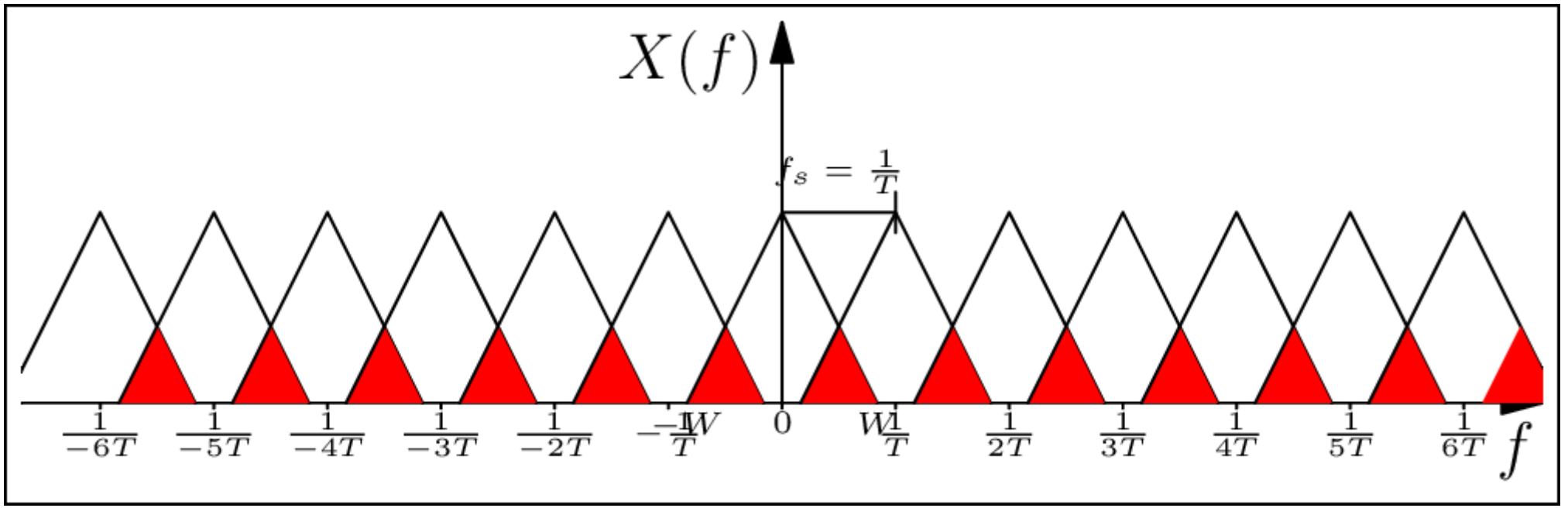
ĐỊNH LÝ LẤY MẪU

$$\omega_s = 2W$$

$$\Rightarrow \omega_s = \frac{2\pi}{T} > 2W$$

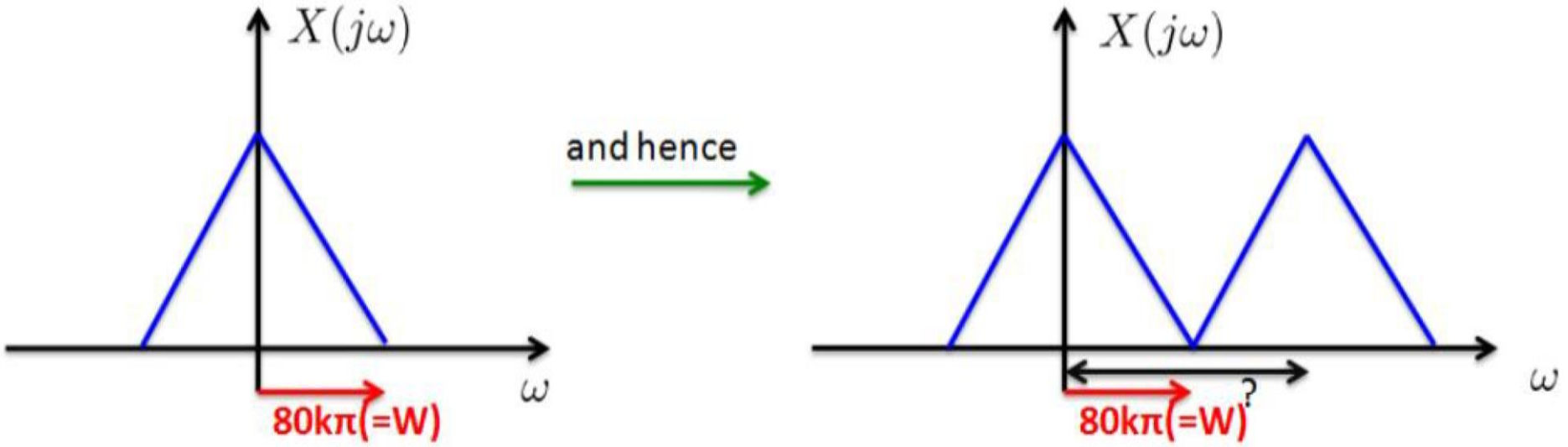




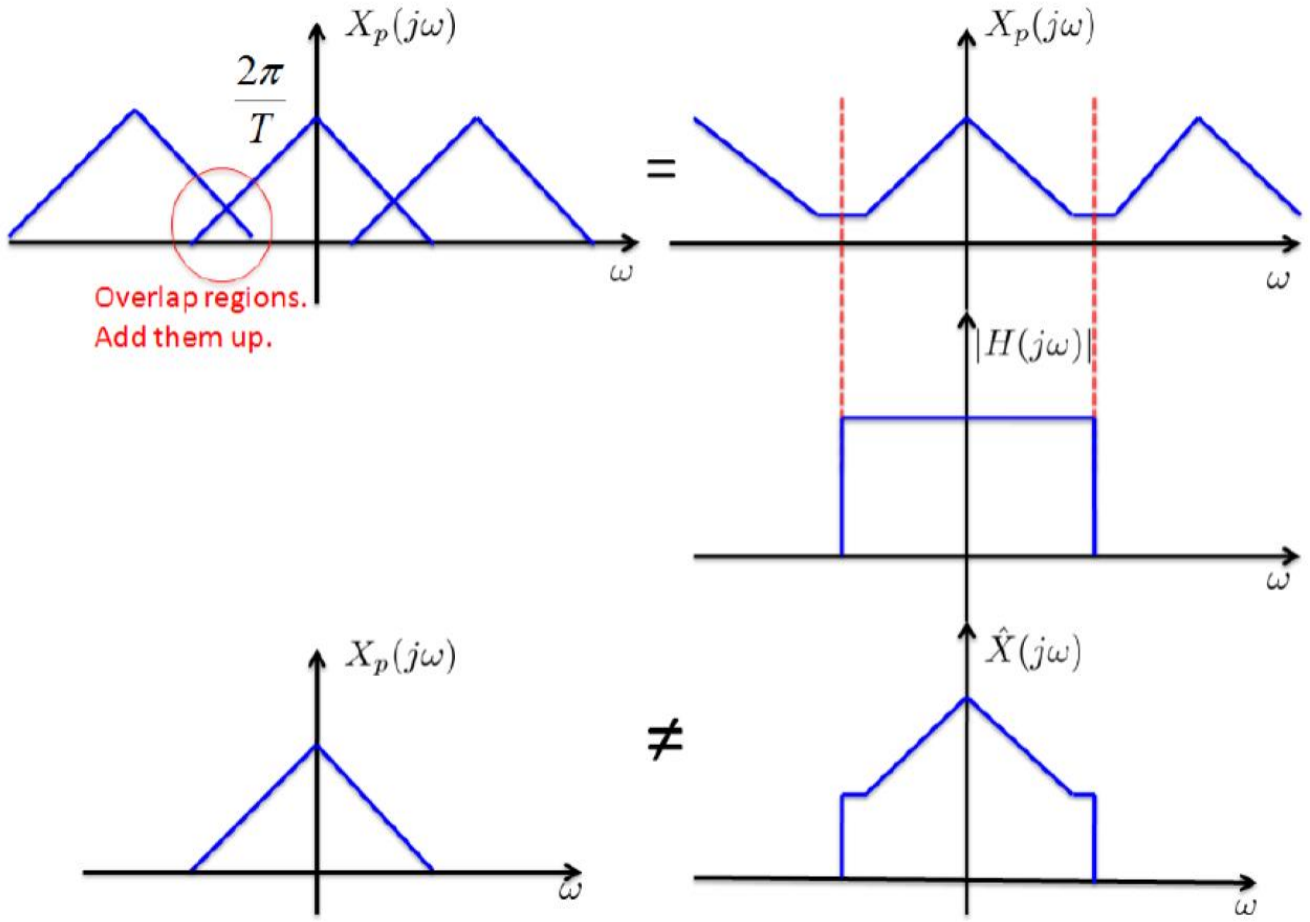


VÍ DỤ

- Nếu tín hiệu có tần số cực đại bằng 40kHz thì tần số lấy mẫu tối thiểu bằng bao nhiêu?



KHÔI PHỤC TÍN HIỆU GỐC TỪ TÍN HIỆU LẤY MẪU



KHÔI PHỤC TÍN HIỆU GỐC TỪ TÍN HIỆU LẤY MẪU

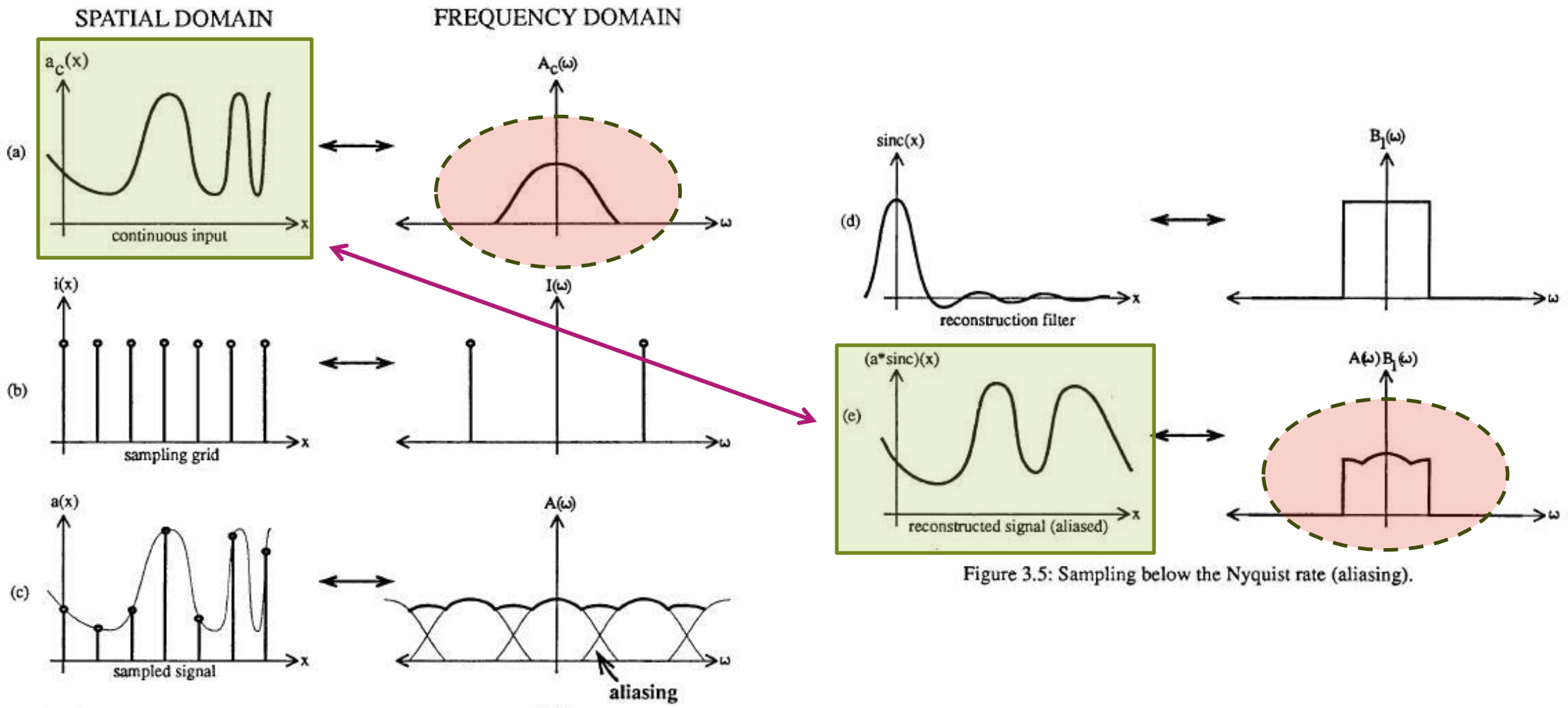


Figure 3.5: Sampling below the Nyquist rate (aliasing).

MATLAB EXAMPLE

Example 1 - Sampling higher than Nyquist rate
Consider the following continuous time signal:

$$x_a(t) = 3\sin(2\pi 0.5t) + 2\sin(2\pi 2.5t)$$

In MATLAB we explore both time-domain and frequency domains by making the following plots:

- **Time domain plots**
- The original time signal $x_a(t)$
- The sampled signal $x[n]$ (discrete time signal) with $f_s = 2.5 f_{\max} = 6.25$ Hz
- The sampled signal superimposed on top of the reconstructed continuous time-signal
- **Frequency domain plots (computed using MATLAB's FFT function)**
- The (double-sided) amplitude spectrum of the orignal time domain signal.
- The amplitude spectrum of the sampled signal.
- The amplitude spectrum of the reconstructed continuous time-signal.

DSP Lab 2 - Sampling and aliasing - Mateo Aboy, Aaron Scher, and Joel Sprunger

MATLAB EXAMPLE

```
clc; clear all; close all;
%%%%%% TIME DOMAIN PLOT (FIGURE 1)
%PLOT THE ORIGINAL CONTINUOUS TIME DOMAIN SIGNAL x(t)
%===============================================
f1=0.5; f2=2.5;  % frequency f1, f2 in Hz
fs=1000*f2; %sampling frequency. When plotting the time-domain signal we
%take a very high sampling frequency so that the plotted signal
%will appear continuous to the eye.
t=[0:1/fs:10]; %plot 10 seconds
x=3*sin(2*pi*f1*t)+2*sin(2*pi*f2*t); %time domain signal we are sampling.
figure('color',[1 1 1]); %Note: Open new figure. Set background color to
white.
subplot(3,1,1)
h=plot(t,x);
ylabel('x(t)','FontSize',16)
xlabel('t [sec]','FontSize',16)
axis tight
set(gca,'FontSize',16)
grid on
```

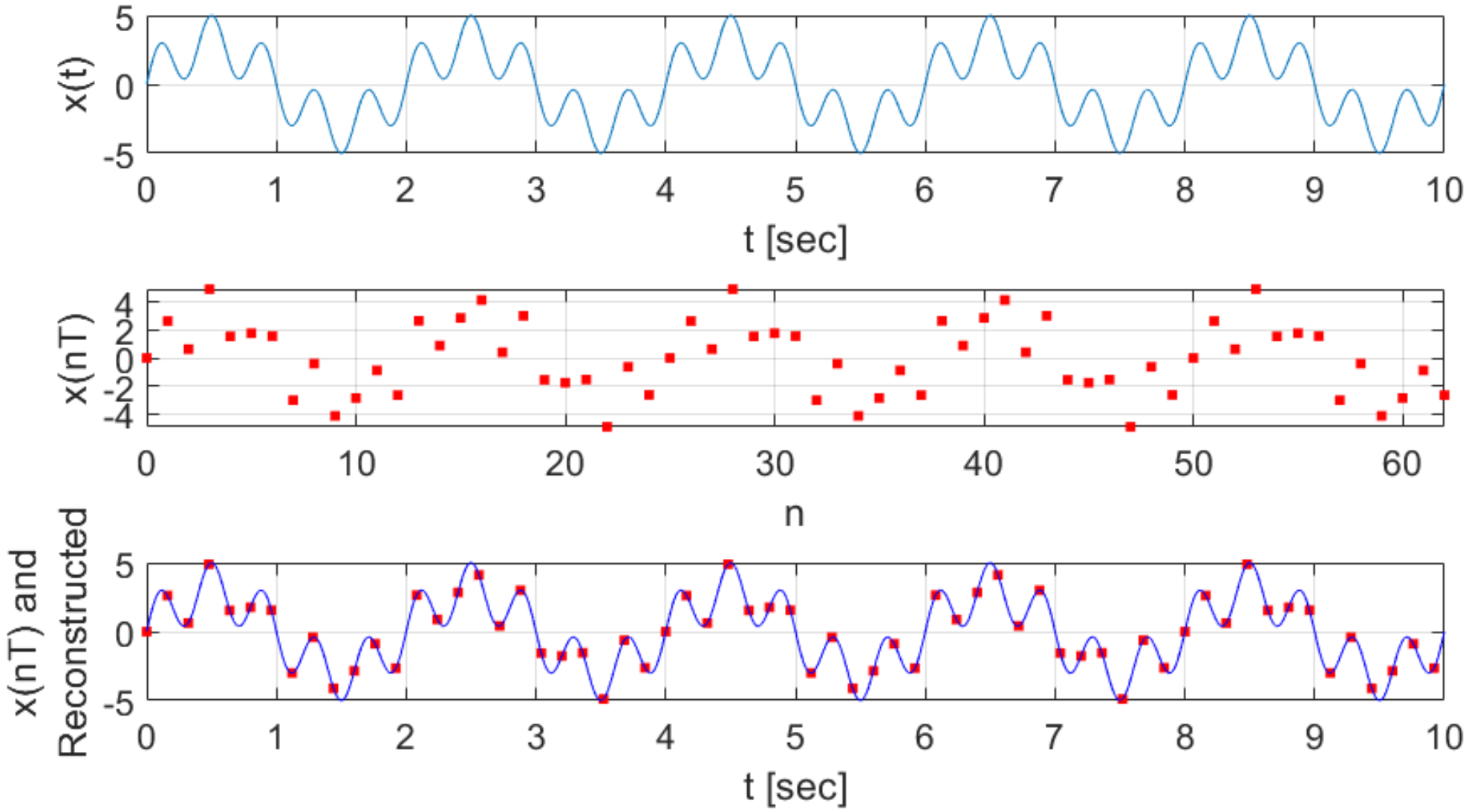
MATLAB EXAMPLE

```
%================================================
%Sampling at fs=2.5*fmax=2.5*f2
%================================================
fs = 2.5*f2; % Sampling Frequency
t_samp=[0:1/fs:10]; %plot 10 seconds.
xs = 3*sin(2*pi*f1*t_samp)+2*sin(2*pi*f2*t_samp); %create sampled signal
subplot(3,1,2);
n_vector=t_samp*fs; %sampled index array: n=[0, 1, 2, 3, ...., N]
h=plot(n_vector,xs,'r.','markers',13);
ylabel('x(nT)','FontSize',16)
xlabel('n','FontSize',16)
axis tight
set(gca,'FontSize',16)
grid on
```

MATLAB EXAMPLE

```
%================================================
%Reconsturcted signal
%================================================
xr=x; %reconstructed signal xr(t) equals the original continuous time
domain signal x(t) because the sampling condition is met
subplot(3,1,3)
h=plot(t_samp,xs,'r.',t,x,'b','markers',13);
str = {'x(nT) and', 'Reconstructed'};
ylabel(str,'FontSize',13)
xlabel('t [sec]','FontSize',16)
axis tight
set(gca,'FontSize',16)
grid on
%
```

MATLAB EXAMPLE



MATLAB EXAMPLE

```
%%%%%% FREQUENCY DOMAIN PLOT (FIGURE 2)
%=============================================
%PLOT SPECTRUM OF ORIGINAL CONTINUOUS TIME DOMAIN SIGNAL x(t)
%=============================================
figure('color',[1 1 1]); %Note: Open new figure. Set background color to white
fs=1000*f2; %sampling frequency. When creating the continuous time-domain
%signal we take a very high sampling frequency so that the plotted signal
%will appear continuous to the eye.
t=[0:1/fs:30]; %plot 30 seconds (for frequency domain plots, you want to
%simulate many periods of the signal so that your frequency domain plots
%have a high resolution (frequency tones appear "sharp")
x=3*sin(2*pi*f1*t)+2*sin(2*pi*f2*t); %time-domain signal
N=2^20; %good general value for FFT (Leave this number alone for Lab 1)
y=fft(x,N); %compute FFT! There is a lot going on "behind the scenes" with
%this one line of code.
z=fftshift(y); %Move the zero-frequency component to the center of the
%array. This is used for finding the double-sided spectrum (as opposed to
%the single-sided spectrum).
```

MATLAB EXAMPLE

```
f_vec=[0:1:N-1]*fs/N-fs/2; %Create frequency vector (this is an array of
%numbers. Each number corresponds to a discrete point in frequency in which we
%shall evaluate and plot the function.)
amplitude_spectrum=abs(z)/length(x); %Extract the amplitude of the spectrum
%Here we also apply a scaling factor of 1/length(x), so that the amplitude
%of the FFT at a frequency component equals that of the ideal double-sided
%spectrum.
subplot(3,1,1)
plot(f_vec,amplitude_spectrum);
set(gca,'FontSize',13) %set font size of axis tick labels to 15
xlabel('Freq [Hz]','fontsize',13)
title('Continuous time spectrum (original signal):','fontsize',17)
%ylabel('Spectrum of x(t)','fontsize',13)
grid on
%axis tight
xlim([-6.25/2,6.25/2])
```

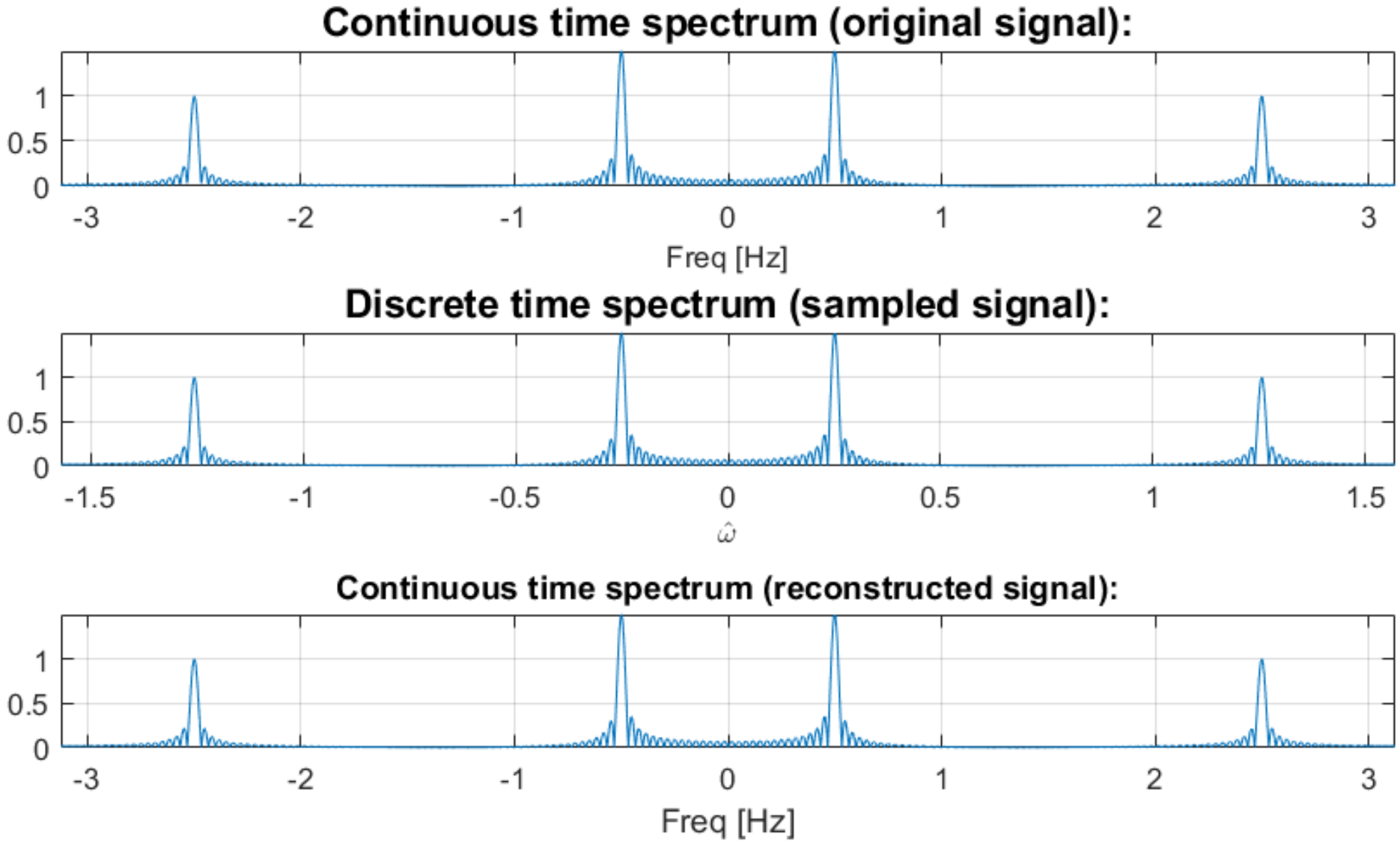
MATLAB EXAMPLE

```
%Sampling at fs=2.5*fmax=2.5*f2
%================================================
fs = 2.5*f2; % Sampling Frequency
t_samp=[0:1/fs:30];
xs = 3*sin(2*pi*f1*t_samp)+2*sin(2*pi*f2*t_samp); %create sampled signal
N=2^20;
y=fft(xs,N);
z=fftshift(y);
amplitude_spectrum=abs(z)/length(xs);
w_vec=[0:1:N-1]*pi/N-pi/2; %create discrete time frequency (omega hat)
subplot(3,1,2)
plot(w_vec,amplitude_spectrum);
set(gca,'FontSize',13) %set font size of axis tick labels to 15
xlabel(['$$\hat{\omega}$$'],'Interpreter','Latex','fontsize',13)
str = {'Spectrum of', 'sampled signal'};
title('Discrete time spectrum (sampled signal):','fontsize',17)
grid on
xlim([-pi/2,pi/2]) %axis tight
```

MATLAB EXAMPLE

```
%================================================
%Reconsturcted signal
%================================================
f_vec=w_vec*fs/pi; %Scale frequency axis
subplot(3,1,3)
plot(f_vec,amplitude_spectrum);
grid on
xlim([-fs/2,fs/2])
xlabel('Freq [Hz]','fontsize',13)
title('Continuous time spectrum (reconstructed signal):','fontsize',17)
set(gca,'FontSize',13) %set font size of axis tick labels to 15
```

MATLAB EXAMPLE



MATLAB EXAMPLE

Example 2 - Sampling **lower** than Nyquist rate
Consider the same continuous time signal as in Ex. 1: $x_a(t) = 3\sin(2\pi 0.5t) + 2\sin(2\pi 2.5t)$

The signal contains two frequency components at $f_1 = 0.5\text{Hz}$ and $f_2 = 2.5\text{Hz}$. We explore the effect of aliasing by sampling $x_a(t)$ at $f_s = 1\text{Hz}$. This sampling frequency does not meet the sampling theorem requirement. Plot in MATLAB:

- **Time domain plots**
- The original time signal $x_a(t)$
- The sampled signal $x[n]$ (discrete time signal) with $f_s = f_{\max}/2.5 = 1$ Hz
- The sampled signal superimposed on top of the original continuous time-signal
- The sampled signal superimposed on top of the reconstructed continuous time-signal
- **Frequency domain plots (computed using MATLAB's FFT function)**
- Similar to Ex. 1 (Note that the reconstructed time-domain signal is **not equal** to the original time-domain signal, because the sampling theorem is not met. The signal is

$$x_r(t) = 3\sin(2\pi 0.5t) + 2\sin(2\pi 0.5t) = 5\sin(2\pi 0.5t)$$

DSP Lab 2 - Sampling and aliasing - Mateo Aboy, Aaron Scher, and Joel Sprunger

MATLAB EXAMPLE

```
clc; clear all; close all;
%%%%%% TIME DOMAIN PLOT (FIGURE 3)
%PLOT THE ORIGINAL CONTINUOUS TIME DOMAIN SIGNAL x(t)
%=============================================
f1=1/2; f2=2.5;  % frequency f1, f2 in Hz
fs=1000*f2; %sampling frequency. When plotting time-domain signal we take a
%very high sampling frequency so that the plotted signal will appear
%continuous to the eye.
t=[0:1/fs:10]; %plot 10 seconds
x=3*sin(2*pi*f1*t)+2*sin(2*pi*f2*t); %time domain signal we are sampling.
figure('color',[1 1 1]); %Open new figure. Set background color to white.
subplot(4,1,1)
h=plot(t,x);
ylabel('x(t)','FontSize',16)
xlabel('t [sec]','FontSize',16)
axis tight
set(gca,'FontSize',16)
grid on
```

MATLAB EXAMPLE

```
%================================================
%Sampling at fs = 1 Hz
%================================================
fs = 1; % Sampling Frequency
t_samp=[0:1/fs:10]; %plot 10 seconds.
xs = 3*sin(2*pi*f1*t_samp)+2*sin(2*pi*f2*t_samp); %create sampled signal
subplot(4,1,2);
n_vector=t_samp*fs; %sampled index array: n=[0, 1, 2, 3, ...., N]
h=plot(n_vector,xs,'r.','markers',15);
ylabel('x(nT)','FontSize',16)
xlabel('n','FontSize',16)
axis([0 10 -5 5])
set(gca,'FontSize',16)
grid on
```

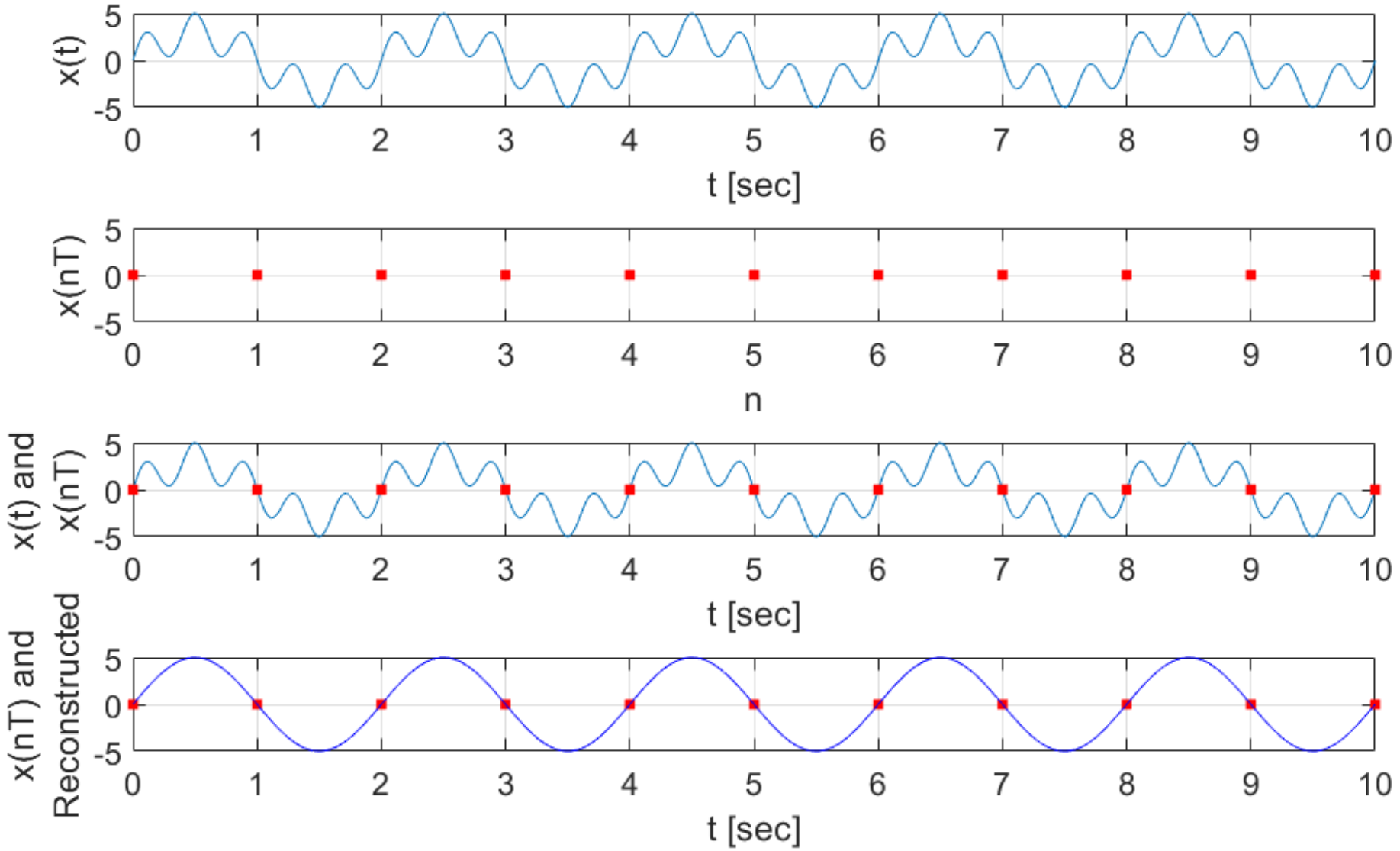
MATLAB EXAMPLE

```
%================================================
%Original and sampled
%================================================
subplot(4,1,3);
n_vector=t_samp*fs; %sampled index array: n=[0, 1, 2, 3, ...., N]
h=plot(t,x,n_vector/fs,xs,'r.','markers',15);
str = {'x(t) and', 'x(nT)'};
ylabel(str,'FontSize',16)
xlabel('t [sec]','FontSize',16)
axis([0 10 -5 5])
set(gca,'FontSize',16)
grid on
```

MATLAB EXAMPLE

```
%================================================
%Reconstructed signal
%================================================
xr=5*sin(2*pi*0.5*t); %reconstructed signal xr(t) does NOT equal the
original continuous time domain signal x(t) because the sampling condition
is not met.
%The reconstructed signal is the corrupted by aliasing.
subplot(4,1,4)
h=plot(t_samp,xs,'r.',t,xr,'b','markers',15);
str = {'x(nT) and', 'Reconstructed'};
ylabel(str,'FontSize',13)
xlabel('t [sec]','FontSize',16)
axis tight
set(gca,'FontSize',16)
grid on
```

MATLAB EXAMPLE



MATLAB EXAMPLE

```
%%%%%% FREQUENCY DOMAIN PLOT (FIGURE 4)
%=============================================
%PLOT SPECTRUM OF ORIGINAL CONTINUOUS TIME DOMAIN SIGNAL x(t)
%=============================================
figure('color',[1 1 1]); %Note: Open new figure. Set background color to
white
fs=1000*f2; %sampling frequency. When creating the continuous time-domain
%signal we take a very high sampling frequency so that the plotted signal
%will appear continuous to the eye.
t=[0:1/fs:30]; %plot 30 seconds (for frequency domain plots, you want to
%simulate many periods of the signal so that your frequency domain plots
%have a high resolution (frequency tones appear "sharp")
x=3*sin(2*pi*f1*t)+2*sin(2*pi*f2*t); %time-domain signal
N=2^20; %good general value for FFT (Leave this number alone for Lab 1)
y=fft(x,N); %compute FFT! There is a lot going on "behind the scenes" with
%this one line of code.
z=fftshift(y); %Move the zero-frequency component to the center of the
%array. This is used for finding the double-sided spectrum
```

MATLAB EXAMPLE

```
f_vec=[0:1:N-1]*fs/N-fs/2; %Create frequency vector (this is an array of
%numbers. Each number corresponds to a discrete point in frequency in which
we
%shall evaluate and plot the function.)
amplitude_spectrum=abs(z)/length(x); %Extract the amplitude of the spectrum
%Here we also apply a scaling factor of 1/length(x), so that the amplitude
%of the FFT at a frequency component equals that of the ideal double-sided
%spectrum.
subplot(3,1,1)
plot(f_vec,amplitude_spectrum);
set(gca,'FontSize',13) %set font size of axis tick labels to 15
xlabel('Freq [Hz]','fontsize',13)
title('Continuous time spectrum (original signal):','fontsize',17)
%ylabel('Spectrum of x(t)','fontsize',13)
grid on
%axis tight
xlim([-6.25/2,6.25/2])
```

MATLAB EXAMPLE

```
%Sampling at fs = 1 Hz
%=============================================
fs = .99; % Sampling Frequency
t_samp=[0:1/fs:30];
xs = 3*sin(2*pi*f1*t_samp)+2*sin(2*pi*f2*t_samp); %sampled signal
N=2^20;
y=fft(xs,N);
z=fftshift(y);
amplitude_spectrum=abs(z)/length(xs);
w_vec=[0:1:N-1]*pi/N-pi/2; %create discrete time frequency (omega hat)
subplot(3,1,2)
plot(w_vec,amplitude_spectrum);
set(gca,'FontSize',13) %set font size of axis tick labels to 15
xlabel(['$$\hat{\omega}$$'],'Interpreter','Latex','fontsize',13)
str = {'Spectrum of', 'sampled signal'};
title('Discrete time spectrum (sampled signal):','fontsize',17)
grid on
xlim([-pi/2,pi/2])
```

MATLAB EXAMPLE

```
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title('Discrete time spectrum (sampled signal):','fontsize',17)
grid on
xlim([-pi/2,pi/2])
```

MATLAB EXAMPLE

```
%================================================
%Reconstructed signal
%================================================
f_vec=w_vec*fs/pi; %Scale frequency axis
subplot(3,1,3)
plot(f_vec,amplitude_spectrum);
grid on
xlim([-fs/2,fs/2])
xlabel('Freq [Hz]','fontsize',13)
title('Continuous time spectrum (reconstructed signal):','fontsize',17)
set(gca,'FontSize',13) %set font size of axis tick labels to 15
```

MATLAB EXAMPLE

